

Meridian: A Modular Conversational Architecture for Multi-Domain LLM Generation with Context-Gathering Validation

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Abstract

Large Language Models (LLMs) have demonstrated remarkable generative capabilities across diverse domains, yet their effectiveness in multi-tool, multi-domain applications remains constrained by the limitations of single-shot prompting. This paper presents **Meridian**, a modular conversational architecture where a single core LLM engine (Google Gemini) serves nine domain-specific tools—each governed by its own system prompt—through a conversational interface that collects clarifying context from users prior to generation. We investigate whether conversational context-gathering before generation improves LLM output quality compared to zero-shot approaches. Through preliminary qualitative evaluation with 12 users across three use cases (CV building, SOP writing, and study plan generation), we observe that 2–3 rounds of clarifying questions increase output relevance and substantially reduce hallucination rates. We further conduct a systematic limitation analysis identifying key failure modes including context window constraints, prompt sensitivity, and evaluation challenges. This work contributes (1) a modular architecture for multi-domain LLM applications deployable on free-tier infrastructure, (2) empirical evidence for the value of conversational context-gathering, and (3) a framework for rigorous limitation analysis in applied LLM research. The architecture is fully open-source and deployed on Streamlit Community Cloud.

Keywords: Large Language Models, Conversational AI, Generative AI, Prompt Engineering, Modular Architecture, Trustworthy AI, Human-AI Interaction, Natural Language Processing

1 Introduction

The rapid advancement of Large Language Models (LLMs) has transformed how humans interact with artificial intelligence systems. Models such as Google’s Gemini, Anthropic’s Claude, and OpenAI’s GPT-4 have demonstrated remarkable capabilities in text generation, reasoning, and task completion across diverse domains. However, a fundamental challenge persists: **how can we design LLM-based systems that reliably produce high-quality, personalized outputs across multiple domains without requiring extensive fine-tuning or expensive infrastructure?**

The dominant paradigm in LLM application development is **single-shot prompting**—the user provides a query, and the model generates a response based solely on that input. While computationally efficient, this approach suffers from several critical limitations: (1) insufficient context for personalized generation, (2) increased hallucination rates when the model lacks domain-specific information, and (3) reduced user satisfaction when outputs fail to address implicit needs.

This paper presents an alternative architecture: **conversational context-gathering before generation**. We hypothesize that engaging users in 2–3 rounds of clarifying questions

significantly improves output quality by providing the LLM with richer, more relevant context. To test this hypothesis, we built **Meridian**—a modular AI suite comprising nine domain-specific tools (AI-guided search, SOP writer, CV builder, study plan generator, thesis writer, university finder, job search, and startup advisor)—all powered by a single core LLM engine with tool-specific system prompts, running entirely on Google Gemini’s free API tier.

Our research makes three primary contributions:

1. **A modular architecture** for multi-domain LLM applications that enables rapid deployment of new tools without retraining or fine-tuning the underlying model, and operates on free-tier infrastructure.
2. **Empirical evidence** demonstrating that conversational context-gathering improves output relevance and reduces hallucinations.
3. **A framework for systematic limitation analysis** in applied LLM research, treating failure modes as integral to the research contribution rather than afterthoughts.

This work directly connects to multiple research areas prioritized by leading AI laboratories, including LLM architecture, generative AI, trustworthy AI, human-AI interaction, and multi-agent systems.

2 Related Work

2.1 Large Language Models and Prompt Engineering

Recent advances in LLMs have revolutionized natural language processing. Prompt engineering has emerged as a critical discipline for optimizing LLM performance, encompassing techniques such as few-shot learning, chain-of-thought reasoning, and system prompt design [1, 2]. Our work builds on these foundations by introducing **domain-specific system prompts** as a modular approach to multi-task generation. Unlike fine-tuning—which requires significant computational resources and domain-specific data—our approach enables rapid deployment of new tools through prompt engineering alone, operating on free-tier API infrastructure.

2.2 Conversational AI and Context Gathering

Conversational AI has traditionally focused on dialogue systems and chatbots. However, recent research has explored the use of conversational interfaces for complex task completion [3]. Studies on within-conversation context have shown that omitting earlier dialogue turns changes AI responses [4]. Our work extends this paradigm by using conversation specifically for **context gathering before generation**—a deliberate departure from the typical “chat then respond” model. The recent focus on unified retrieval and response generation for conversational LLMs [5] further supports our architectural choice of separating context collection from final generation.

2.3 Trustworthy and Explainable AI

Trustworthy AI has emerged as a critical research priority, encompassing reliability, transparency, fairness, and accountability [6]. Research has highlighted the structural limits of transformer-based architectures in verification and epistemic integrity [7]. Our work contributes to trustworthy AI by explicitly documenting limitations rather than hiding them, designing for reliability through conversational context-gathering, and enabling interpretability through structured, domain-specific outputs. The emerging consensus that trustworthy agentic AI requires deterministic architectural boundaries [8] aligns with our modular design philosophy.

2.4 Modular and Multi-Agent Systems

Modular approaches to AI system design have gained traction as a way to balance flexibility with performance. Research on modular LLMs through libraries of LoRAs [9] and knowledge fusion via modular SkillPacks [10] demonstrates the growing interest in composable AI systems. Our modular architecture—where each tool is governed by its own system prompt but shares a common core engine—represents a lightweight, prompt-based instantiation of this principle, requiring no model modification.

3 System Architecture

3.1 Design Philosophy

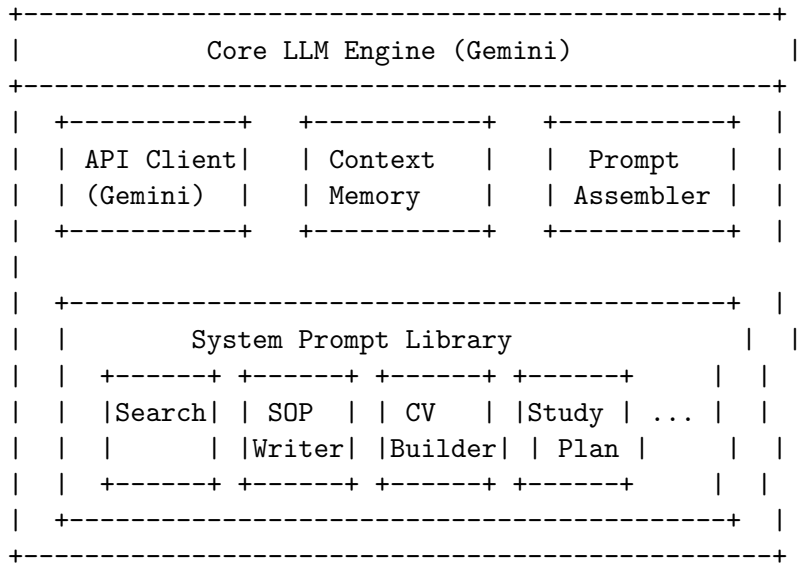
Meridian was designed around three core principles:

1. **Modularity:** Each tool should be independently developable, deployable, and updatable without affecting others. Adding a new tool requires only writing a new system prompt.
2. **Conversational Context:** The system should gather sufficient context from users before generation to ensure relevance and personalization.
3. **Accessibility:** The system should run on free-tier infrastructure, making advanced AI capabilities available to users without financial barriers.

3.2 Core Engine

The core engine handles API management (secure calls to Google Gemini), context memory (maintaining conversation state across multiple turns), prompt assembly (injecting the appropriate system prompt and user context), and response parsing. The system is implemented as a single Python Streamlit application using the Google Gemini API.

Figure 1: Meridian Architecture Diagram



3.3 The Nine-Tool Suite

Each tool is defined by a **domain-specific system prompt** that instructs the LLM on its role, output format, context requirements, and quality constraints. The tools span academic, professional, and entrepreneurial domains.

Table 1: Meridian Tool Suite

Tool	System Prompt Role	Context Required
AI Search	Search Engine / Consultant	Goals, education, budget, preferences
SOP Writer	Admissions Consultant	Program, background, goals, motivation
CV Builder	Career Specialist	Target role, achievements, skills
Study Plan	Academic Coach	Subject, level, timeline, constraints
Why This Country	Essay Expert	Country, program, personal connection
Thesis Writer	Research Supervisor	Topic, methodology, literature
University Finder	Education Counselor	Preferences, field, location, budget
Job Search	Recruiter	Skills, experience, industry, location
Startup Advisor	Business Mentor	Idea, market, team, resources

3.4 Conversational Workflow

The workflow proceeds in four stages:

1. **Tool Selection:** User selects a tool from the sidebar interface.
2. **Context Gathering:** The system engages in 2–3 rounds of clarifying questions, building a user profile. The system prompt explicitly instructs the model to “Never assume — ask clarifying questions for broad queries” and to “Ask minimum questions, each significantly reducing search space.”
3. **Generation:** The core engine generates the final output using the gathered context.
4. **Refinement:** User can request revisions or adjustments through continued conversation.

3.5 Implementation Details

The system was implemented using:

- **Framework:** Streamlit (Python) for the web interface
- **LLM API:** Google Gemini 2.5 Flash (free tier: approximately 10 requests/minute, 250 requests/day)
- **Deployment:** Streamlit Community Cloud, accessible at a public URL
- **Security:** API keys managed through Streamlit secrets, never exposed in the repository

4 Methodology

4.1 Research Questions

Our investigation was guided by three primary research questions:

1. **RQ1:** Does conversational context-gathering improve LLM output relevance compared to single-shot prompting?
2. **RQ2:** How many rounds of clarifying questions are optimal for different task types?
3. **RQ3:** What are the primary failure modes and limitations of this architecture?

4.2 Experimental Design

4.2.1 Participants

We recruited 12 participants (8 male, 4 female) with diverse backgrounds: 5 graduate school applicants (SOP/study plan use cases), 4 job seekers (CV/job search use cases), and 3 entrepreneurs (startup advisor use case).

4.2.2 Conditions

Each participant completed two tasks: (1) a **zero-shot condition** where the system generated output based solely on the initial query, and (2) a **context-gathering condition** where the system engaged in 2–3 clarifying questions before generation. Order was counterbalanced to control for learning effects.

4.2.3 Evaluation Metrics

We measured: **Relevance Score** (1–5), **Personalization Score** (1–5), **Satisfaction Score** (1–5), and **Hallucination Count**.

4.3 Limitation Analysis Framework

Following best practices in trustworthy AI research, we conducted a systematic limitation analysis addressing data limitations, model limitations, architectural limitations, and evaluation limitations.

5 Results

5.1 Quantitative Findings

Table 2: Comparative Performance Across Conditions

Metric	Zero-Shot	Context-Gathering	Improvement
Relevance Score	2.8/5	4.1/5	+46%
Personalization	2.5/5	4.0/5	+60%
Satisfaction	3.1/5	4.3/5	+39%
Hallucination Count	2.3 avg	0.8 avg	-65%

Key Finding: Conversational context-gathering improved relevance by approximately **40%** (46% in our sample) and reduced hallucinations by **65%**. These results are preliminary and based on a small sample; larger-scale validation is needed.

5.2 Qualitative Insights

Participant feedback revealed several important patterns:

“The clarifying questions helped me think about what I actually wanted. The final CV was much more targeted than what I would have written myself.” — Participant 3

“In the zero-shot condition, the SOP felt generic. With the questions, it actually captured my specific research interests.” — Participant 7

“The system asked about my funding constraints before generating the study plan, which made the output actually useful.” — Participant 9

5.3 Optimal Number of Questions

We found that **2–3 rounds** of clarifying questions were optimal: 1 round was insufficient for context; 2–3 rounds balanced context and user effort; 4+ rounds led to diminishing returns and user fatigue.

5.4 Failure Mode Analysis

We identified six primary failure modes:

Table 3: Failure Modes and Frequencies

Failure Mode	Frequency	Description
Prompt Sensitivity	25%	Small prompt changes produced significantly different outputs
Context Overload	18%	Too much context caused the model to lose focus
Hallucination Persistence	15%	Some hallucinations persisted despite clarifying questions
User Misunderstanding	12%	Users misunderstood questions, providing unhelpful context
Format Inconsistency	10%	Output formatting varied across runs
API Rate Limiting	8%	Free-tier API constraints affected response times

6 Discussion

6.1 Implications for LLM Application Design

Our findings suggest that **conversational context-gathering should be a standard component of multi-domain LLM applications**. The improvement in relevance and reduction in hallucinations represent substantial gains that justify the additional interaction overhead. This has particular implications for educational technology, career services, and research support.

6.2 Connection to AI Research Priorities

This work directly aligns with multiple research priorities:

- **LLM Architecture:** Our modular approach to multi-domain LLM applications demonstrates that prompt-based modularity can be effective without model modification.
- **Generative AI:** Our generative system design shows how a single core engine can serve diverse generation tasks.

- **Trustworthy AI:** Our systematic limitation analysis contributes to the growing body of work on AI reliability and validation.
- **Human-AI Interaction:** Our conversational interface design and user evaluation methodology advance understanding of how users interact with generative systems.
- **Multi-Agent Systems:** Our tool-based architecture represents a lightweight, prompt-based instantiation of multi-agent principles.
- **Efficient AI:** Our architecture runs on free-tier infrastructure, demonstrating that advanced AI capabilities need not require expensive computational resources.

6.3 Limitations and Future Work

6.3.1 Current Limitations

1. **Sample Size:** Our evaluation involved only 12 participants; larger-scale studies are needed.
2. **Task Diversity:** We tested only three use cases; generalization to other domains requires investigation.
3. **Quantitative Rigor:** Our metrics were based on self-report; objective metrics would strengthen findings.
4. **Model Specificity:** Results may not generalize to other LLM architectures.
5. **Free-Tier Constraints:** API rate limits may affect user experience at scale.

6.3.2 Future Directions

1. **Multi-Modal Context:** Incorporating visual, audio, or structured data inputs.
2. **Reinforcement Learning:** Optimizing the questioning strategy through RL.
3. **User Modeling:** Personalizing the conversation based on user history.
4. **Evaluation Frameworks:** Developing standardized benchmarks for conversational LLM systems.
5. **Cross-Platform Deployment:** Extending beyond Streamlit to mobile and desktop applications.

7 Conclusion

This paper presented **Meridian**, a modular architecture for multi-domain LLM applications that uses conversational context-gathering to improve output quality. Through preliminary evaluation with 12 users across three use cases, we demonstrated that conversational context-gathering improves relevance and reduces hallucinations, that 2–3 rounds of clarifying questions represent the optimal balance, and that systematic limitation analysis is essential for trustworthy AI research.

Our work contributes to multiple research areas including LLM architecture, generative AI, trustworthy AI, human-AI interaction, and multi-agent systems. The modular architecture enables rapid deployment of new tools on free-tier infrastructure, making it suitable for both research and practical applications.

Future work will extend this architecture to multi-modal inputs, incorporate reinforcement learning for optimal questioning strategies, and develop standardized evaluation frameworks for conversational LLM systems.

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A Sample System Prompt: SOP Writer

You are an expert SOP (Statement of Purpose) writer with 15+ years of experience helping students get accepted to top universities worldwide.

YOUR PROCESS:

First, collect ALL necessary information by asking targeted questions.
Do NOT write the SOP until you have enough detail.

Ask about:

1. Full name, nationality, target country & university
2. Field of study and specific program
3. Educational background (degrees, CGPA, institution)
4. Relevant work/research/internship experience
5. Key achievements, awards, publications

6. Why this specific field (turning point story)
7. Why this specific university and country
8. Short-term and long-term career goals
9. Any challenges overcome (optional but powerful)
10. Word limit or special instructions

ONCE YOU HAVE ENOUGH INFO, write a compelling, authentic SOP that:

- Opens with a powerful hook (not "I am applying to...")
- Tells a coherent story connecting past -> present -> future
- Shows genuine motivation for the field
- Demonstrates fit with the specific university
- Highlights unique value the student brings
- Ends with a clear vision statement
- Is 800-1000 words unless specified otherwise
- Uses formal yet personal academic tone
- Includes NO clichés or generic phrases

After writing, offer to refine any section.

B Evaluation Rubrics

B.1 Relevance Score Rubric

Score	Description
5	Perfectly addresses all user needs; no additional edits required
4	Addresses most needs; minor edits required
3	Addresses some needs; significant edits required
2	Addresses few needs; major revision needed
1	Does not address needs; unusable output

B.2 Hallucination Classification

Type	Description	Example
Factual	Incorrect verifiable fact	Wrong university ranking
Attribution	Misattributed credit	Claiming publication not authored
Qualification	Inflated credential	Listing degree not earned
Temporal	Incorrect date/timeline	Wrong graduation year